Prerequisite Quiz

EE 3223: C++ and Data Structures

Name: Ivan Garcia **Date:** 5/28/26

This is graded but you'll receive full credit as long as you complete it (even if you get a question incorrect). Try to work through it closed-book - no compiler, no notes, no internet.

Time: about 15 minutes.

Part A - Types, operators, and expressions

1. What does this print?

```
int a = 5, b = 2;
double c = a / b;
cout << c;
```

- (a) 2.5
- (b) 2
- (c) 2.0
- (d) 3

2. What does this print?

```
cout << 2 + 3 * 4;
```

- (a) 20
- (b) 14
- (c) 24
- (d) 9

Part B - Control flow

3. What does this print when score is 72?

```
int score = 72;
if (score >= 90)      cout << "A";
else if (score >= 80) cout << "B";
else if (score >= 70) cout << "C";
else                  cout << "F";
```

- (a) A
- (b) B
- (c) C
- (d) F

4. What does this print? (Note the missing break statements.)

```
int x = 2;
switch (x) {
    case 1: cout << "a";
    case 2: cout << "b";
    case 3: cout << "c"; break;
    default: cout << "d";
}
```

- (a) b
- (b) bc
- (c) abc
- (d) bcd

Part C - Functions and scope

5. What does this print?

```
void addTen(int n) { n = n + 10; }
int main() {
    int x = 5;
    addTen(x);
    cout << x;
}
```

- (a) 15
- (b) 5
- (c) 10
- (d) 0

6. (Short answer - write code.) Complete the function so it returns the larger of its two arguments:

```
int max2(int a, int b) {
    // your code here
}
```

Your answer:

```
if (a > b)  cout << a;
else if (b > a)  cout << b;
else if (a == b) cout << a;
```

Part D - Arrays

7. Given int a[5]; which indices are valid for reading a[i]?

- (a) 1 through 5
- (b) 0 through 5
- (c) 0 through 4
- (d) 1 through 4

8. What does this print?

```
int a[] = {4, 9, 2, 7};
int m = a[0];
for (int i = 1; i < 4; i++)
    if (a[i] > m) m = a[i];
cout << m;
```

- (a) 4
- (b) 22
- (c) 7
- (d) 9

Part E - Tracing

9. What does this print?

```
for (int i = 1; i <= 3; i++) {
    for (int j = 1; j <= i; j++)
        cout << "*";
    cout << "\n";
}
```

- (a) three lines: *, **, ***
- (b) three lines: ***, **, *
- (c) one line: ******
- (d) three lines of ***

Part F - Number sense

10. The 4-bit binary number 1011 equals what in decimal?

- (a) 9
- (b) 11
- (c) 13
- (d) 1011